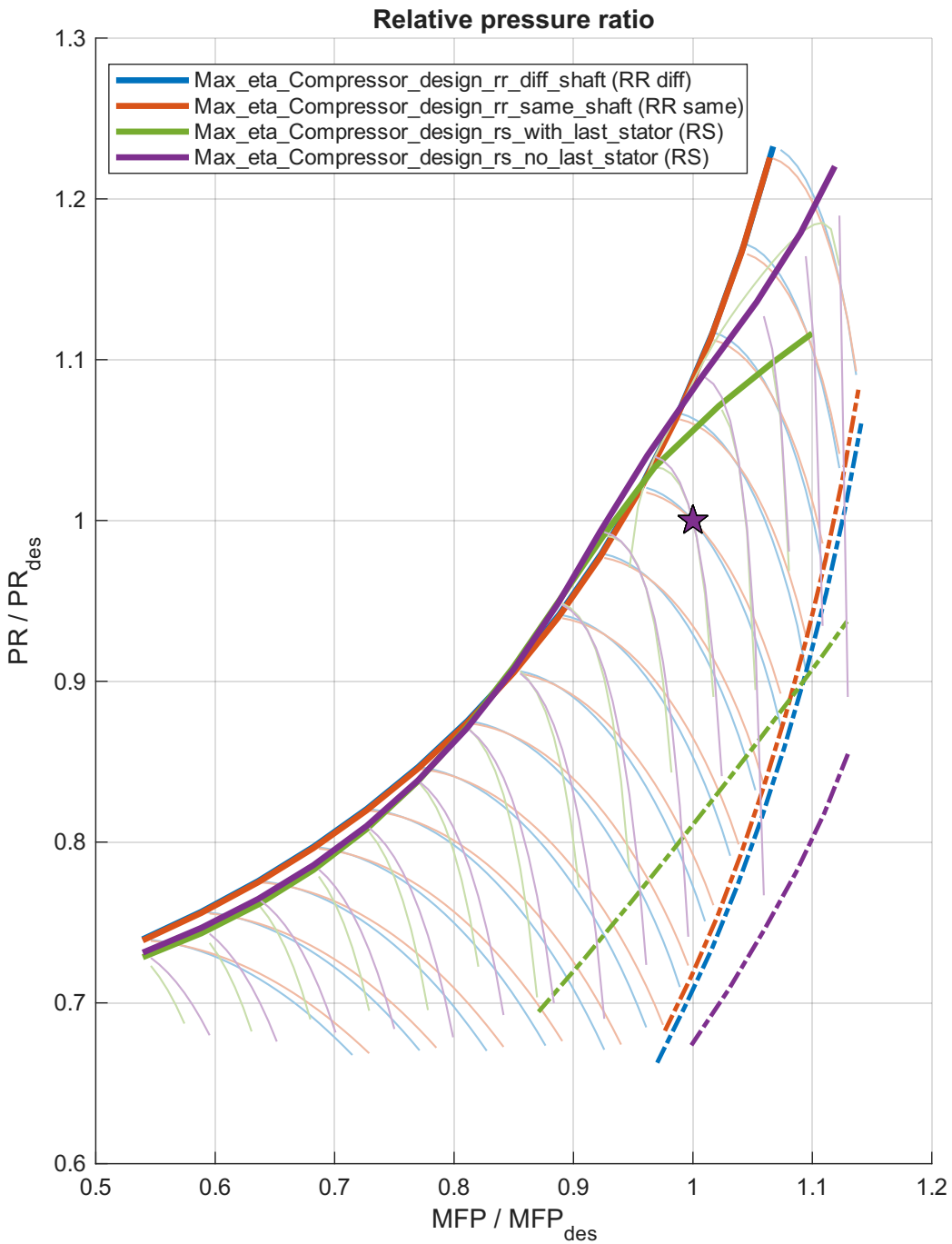
Comparison

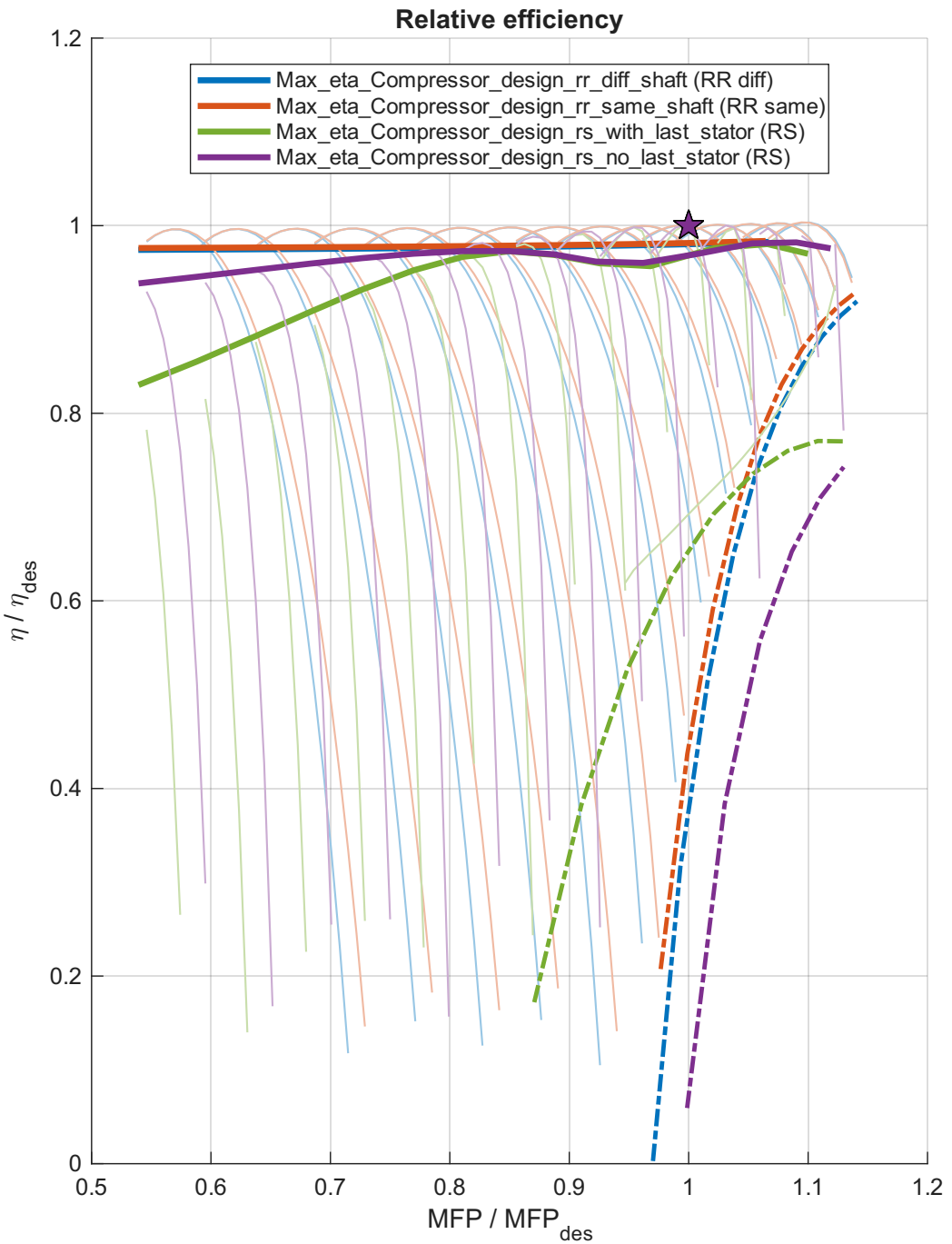
Runs:
  1. Max_eta_Compressor_design_rr_diff_shaft (RR diff) - Rotor-Rotor different shaft - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_eta_Compressor_design_rr_diff_shaft.mat
  2. Max_eta_Compressor_design_rr_same_shaft (RR same) - Rotor-Rotor same shaft - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_eta_Compressor_design_rr_same_shaft.mat
  3. Max_eta_Compressor_design_rs_with_last_stator (RS) - Rotor-Stator - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_eta_Compressor_design_rs_with_last_stator.mat
  4. Max_eta_Compressor_design_rs_no_last_stator (RS) - Rotor-Stator - E:\books_IMP\TurboMachinary\Project\Gas_Turbomachinery_Project_Team11\App\Research_files\Max_eta_Compressor_design_rs_no_last_stator.mat

Line styles on the graphs: solid = first stability limit, dash-dot = choke line, star = design point, light lines = speed lines.

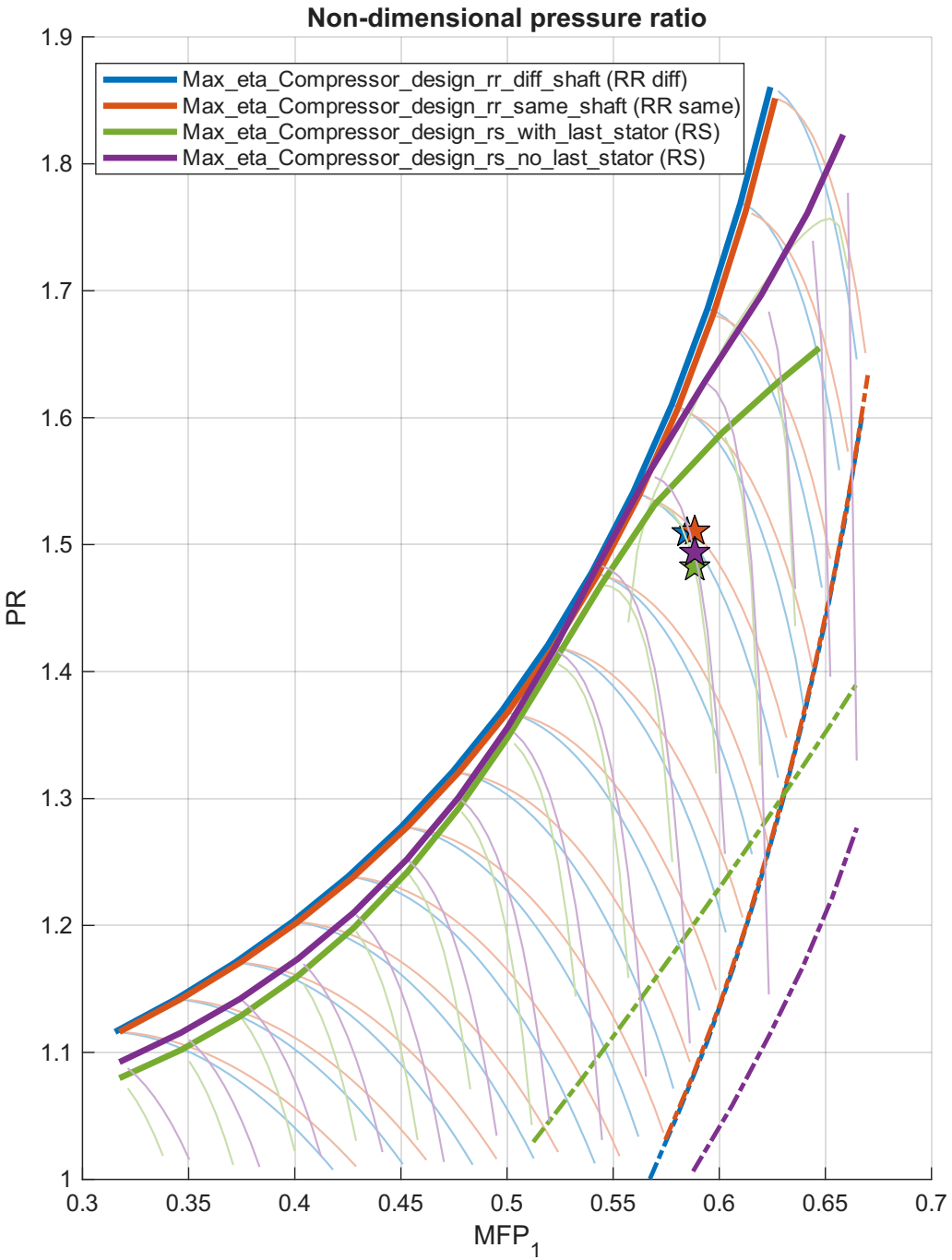
Label	Configuration	Source	PR_des	eta_des	First limit (N=1)	SM [%]	Choke margin [%]	Range [%]	Koch ratio (max)	Koch applicability	U/(2a_p)
Max_eta_Compressor_design_rr_diff_s...	Rotor-Rotor different shaft	Max_eta_Compressor_design_rr_diff_s...	1.5094	0.9589	stall onset	6.7	7.7	11.1	1.261	not recommended	0.404
Max_eta_Compressor_design_rr_same_s...	Rotor-Rotor same shaft	Max_eta_Compressor_design_rr_same_s...	1.5106	0.9586	stall onset	6.6	7.8	11.4	1.313	not recommended	0.378
Max_eta_Compressor_design_rs_with_l...	Rotor-Stator	Max_eta_Compressor_design_rs_with_l...	1.4824	0.9123	peak	6.7	2.1	5.2	1.036	indicative	0.377
Max_eta_Compressor_design_rs_no_las...	Rotor-Stator	Max_eta_Compressor_design_rs_no_las...	1.4939	0.9312	peak	8.2	3	6.6	1.036	indicative	0.377

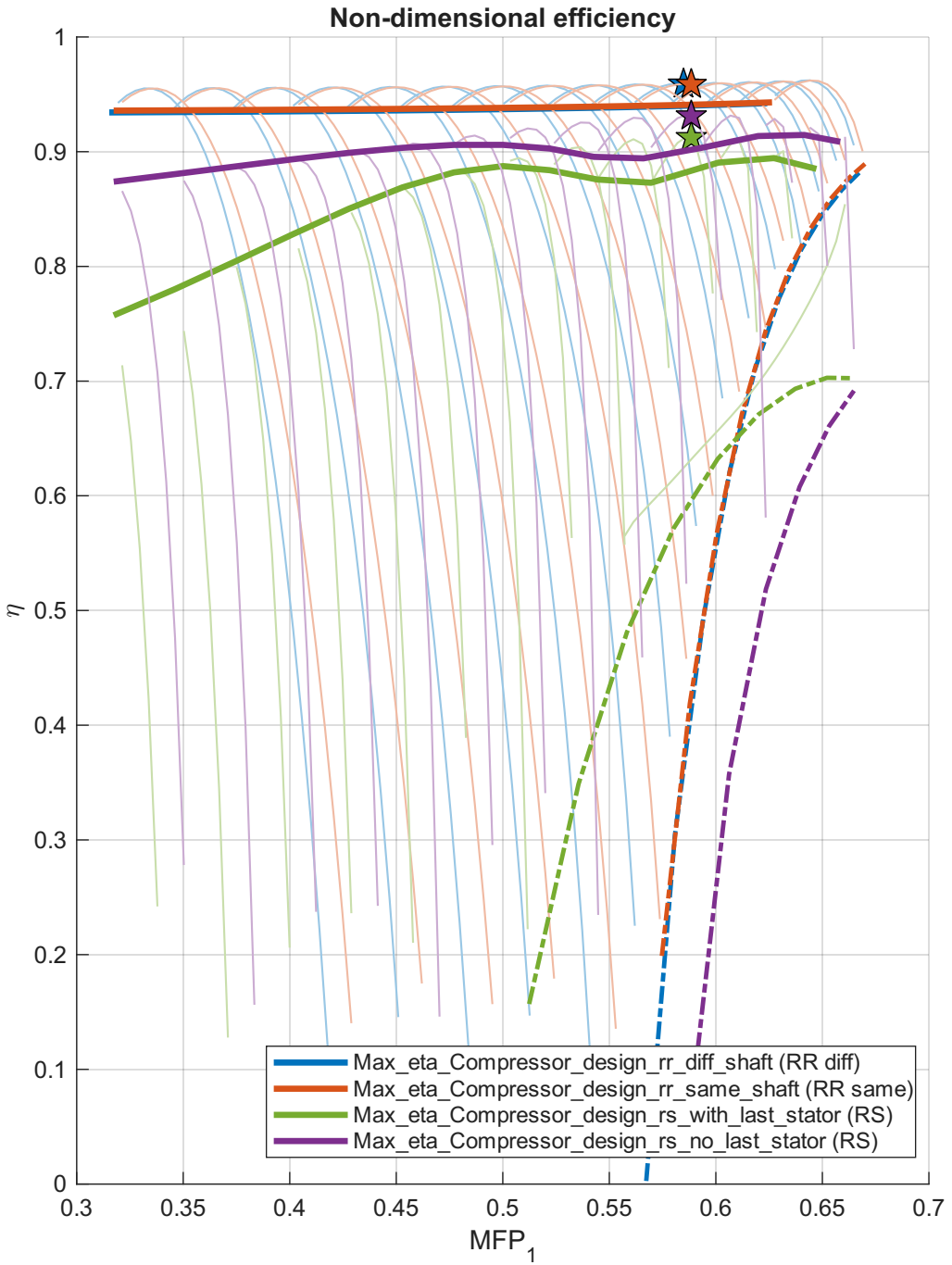
Relative comparison





NonDimensional comparison





Inputs: Max_eta_Compressor_design_rr_diff_shaft (RR diff)

Variable	Value	Unit	Source
Configuration	Rotor-Rotor different shaft		
Input source	.mat: Max_eta_Compressor_design_rr_...		
alpha1	34.0366	deg	.mat: Max_eta_Compressor_design_rr_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rr_...
A_vec	0.00296617 0.00273918 0.00265316	m^2	.mat: Max_eta_Compressor_design_rr_...
beta1_dash	53.5959	deg	.mat: Max_eta_Compressor_design_rr_...
beta2_dash	21.4107	deg	.mat: Max_eta_Compressor_design_rr_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rr_...
w_loss_rotor	0.0342878	-	.mat: Max_eta_Compressor_design_rr_...
beta3_dash	35.0314	deg	.mat: Max_eta_Compressor_design_rr_...
beta4_dash	0.389282	deg	.mat: Max_eta_Compressor_design_rr_...
deviation_1	12.6259	deg	.mat: Max_eta_Compressor_design_rr_...
deviation_2	11.1621	deg	.mat: Max_eta_Compressor_design_rr_...
incidence_1	-6.42119	deg	.mat: Max_eta_Compressor_design_rr_...
incidence_2	-4.58787	deg	.mat: Max_eta_Compressor_design_rr_...
deflection_1	13.1381	deg	.mat: Max_eta_Compressor_design_rr_...
deflection_2	18.8922	deg	.mat: Max_eta_Compressor_design_rr_...
w_loss_rotor2	0.0286249	-	.mat: Max_eta_Compressor_design_rr_...
U1	293.36	m/s	.mat: Max_eta_Compressor_design_rr_...
U2	278.692	m/s	.mat: Max_eta_Compressor_design_rr_...
sigma_1	0.507386	-	.mat: Max_eta_Compressor_design_rr_...
sigma_2	0.500805	-	.mat: Max_eta_Compressor_design_rr_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rr_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_eta_Compressor_design_rr_diff_shaft (RR diff)

```
Rotor-Rotor Different Shaft  |  PR_des = 1.5094, eta_des = 0.9589  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = stall onset at m = 0.4783 kg/s (PR 1.541); stall margin SM = 6.7 %
   peak of the speed line (zero slope): m = 0.4741 kg/s | Howell stall onset: m = 0.4783 kg/s
   Koch stage limit: m = 0.5142 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5383 kg/s (Rotor 1 exit (relative)) -> choke margin 7.7 %, operating range (m_choke - m_lim)/m_choke = 11.1 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.404 -> surge if the installation has V_p/(A_c L_c) > 3.92 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (stall onset): highest loss rise Rotor 1 (omega/omega_des = 2.00), lowest de Haller Rotor 1 (0.77). Details: Tables >
      Blade rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 15 of 15 speed lines (N/N_des 0.50 to 1.20).
   why: Rotors R1+R2: L/g2 = 0.54 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: R2: effective dynamic-head factor F_ef = 0.25, outside 0.39-1.11. The upstream wakes enter in a frame turning the same way at a similar
      speed; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: BEYOND THE LIMIT AT DESIGN (1.26)
     - RAISE the solidity x2.09 (stage L/g2 0.54 -> 1.12): R1 0.51 -> 1.06, R2 0.50 -> 1.05; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 33% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 1.16
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 1.18
     - R2: effective dynamic-head factor F_ef = 0.25 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
      1 the ratio would be 0.91 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: L/g2 = 0.54 (data 0.6-2.2), F_ef = [1.04 0.25] (data 0.39-1.11) -> indicative only
```

Inputs: Max_eta_Compressor_design_rr_same_shaft (RR same)

Variable	Value	Unit	Source
Configuration	Rotor-Rotor same shaft		
Input source	.mat: Max_eta_Compressor_design_rr_...		
alpha1	30.3676	deg	.mat: Max_eta_Compressor_design_rr_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rr_...
A_vec	0.00283128 0.00262365 0.00253207	m^2	.mat: Max_eta_Compressor_design_rr_...
beta1_dash	50.3693	deg	.mat: Max_eta_Compressor_design_rr_...
beta2_dash	18.077	deg	.mat: Max_eta_Compressor_design_rr_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rr_...
w_loss_rotor	0.032576	-	.mat: Max_eta_Compressor_design_rr_...
beta3_dash	34.9249	deg	.mat: Max_eta_Compressor_design_rr_...
beta4_dash	-0.0569908	deg	.mat: Max_eta_Compressor_design_rr_...
deviation_1	12.2907	deg	.mat: Max_eta_Compressor_design_rr_...
deviation_2	11.1084	deg	.mat: Max_eta_Compressor_design_rr_...
incidence_1	-6.04865	deg	.mat: Max_eta_Compressor_design_rr_...
incidence_2	-4.55729	deg	.mat: Max_eta_Compressor_design_rr_...
deflection_1	13.953	deg	.mat: Max_eta_Compressor_design_rr_...
deflection_2	19.3162	deg	.mat: Max_eta_Compressor_design_rr_...
w_loss_rotor2	0.0293003	-	.mat: Max_eta_Compressor_design_rr_...
U	274.775	m/s	.mat: Max_eta_Compressor_design_rr_...
sigma_1	0.500466	-	.mat: Max_eta_Compressor_design_rr_...
sigma_2	0.512047	-	.mat: Max_eta_Compressor_design_rr_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rr_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_eta_Compressor_design_rr_same_shaft (RR same)

```
Rotor-Rotor Same Shaft  |  PR_des = 1.5106, eta_des = 0.9586  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = stall onset at m = 0.4776 kg/s (PR 1.538); stall margin SM = 6.6 %
   peak of the speed line (zero slope): m = 0.4740 kg/s | Howell stall onset: m = 0.4776 kg/s
   Koch stage limit: m = 0.5169 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5389 kg/s (Rotor 1 exit (relative)) -> choke margin 7.8 %, operating range (m_choke - m_lim)/m_choke = 11.4 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.378 -> surge if the installation has V_p/(A_c L_c) > 4.47 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (stall onset): highest loss rise Rotor 1 (omega/omega_des = 2.00), lowest de Haller Rotor 1 (0.78). Details: Tables >
      Blade rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): NOT RECOMMENDED
   Not recommended for this design: it is outside the basis of Koch's correlation (see why). The Koch line is hidden by default; use Howell, the peak
      and the Blade rows table instead.
   Koch line: drawn on 15 of 15 speed lines (N/N_des 0.50 to 1.20).
   why: Rotors R1+R2: L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: R2: effective dynamic-head factor F_ef = 0.25, outside 0.39-1.11. The upstream wakes enter in the same rotating frame (same shaft), so they
      are not re-energized; Koch's correlation does not cover this.
   why: Rotor-rotor stage: Koch's database holds rotor-stator stages only.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Rotors R1+R2: BEYOND THE LIMIT AT DESIGN (1.31)
     - RAISE the solidity x2.27 (stage L/g2 0.53 -> 1.20): R1 0.50 -> 1.13, R2 0.51 -> 1.16; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 35% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 1.21
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 1.23
     - R2: effective dynamic-head factor F_ef = 0.25 (upstream wakes enter in the same or a nearly equal frame and are not re-energized); with F_ef =
      1 the ratio would be 0.91 -> a stator between the rotors or a larger frame-speed difference (e.g. counter-rotation) helps
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2), F_ef = [0.99 0.25] (data 0.39-1.11) -> indicative only
```

Inputs: Max_eta_Compressor_design_rs_with_last_stator (RS)

Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_eta_Compressor_design_rs_...		
alpha1	30.3676	deg	.mat: Max_eta_Compressor_design_rs_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rs_...
A_vec	0.00283128 0.00262358 0.00243314 0....	m^2	.mat: Max_eta_Compressor_design_rs_...
beta1_dash	50.3693	deg	.mat: Max_eta_Compressor_design_rs_...
beta2_dash	18.077	deg	.mat: Max_eta_Compressor_design_rs_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rs_...
w_loss_rotor	0.0324827	-	.mat: Max_eta_Compressor_design_rs_...
deviation	12.2907	deg	.mat: Max_eta_Compressor_design_rs_...
incidence	-6.04865	deg	.mat: Max_eta_Compressor_design_rs_...
deflection	13.953	deg	.mat: Max_eta_Compressor_design_rs_...
w_loss_stator	0.0266822	-	.mat: Max_eta_Compressor_design_rs_...
U	274.775	m/s	.mat: Max_eta_Compressor_design_rs_...
sigma_rotor	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
sigma_stator	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rs_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_eta_Compressor_design_rs_with_last_stator (RS)

```
Rotor-Stator  |  PR_des = 1.4824, eta_des = 0.9123  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = peak at m = 0.4840 kg/s (PR 1.532); stall margin SM = 6.7 %
   peak of the speed line (zero slope): m = 0.4840 kg/s | Howell stall onset: m = 0.4825 kg/s
   Koch stage limit: m = 0.5018 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5106 kg/s (Stator 2 exit (absolute)) -> choke margin 2.1 %, operating range (m_choke - m_lim)/m_choke = 5.2 %
WARNING: N/N_des = 1.20 has no stable range (none (beyond onset)).
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.377 -> surge if the installation has V_p/(A_c L_c) > 4.51 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 2 (omega/omega_des = 1.92), lowest de Haller Stator 2 (0.79). Details: Tables > Blade
      rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: drawn on 6 of 15 speed lines (N/N_des 0.85 to 1.10).
   why: Stage 1 (R1+S1): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2+S2): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): BEYOND THE LIMIT AT DESIGN (1.04)
     - RAISE the solidity x1.42 (stage L/g2 0.53 -> 0.76): R1 0.50 -> 0.71, S1 0.50 -> 0.71; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 18% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.95
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.97
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
  Stage 2 (R2+S2): BEYOND THE LIMIT AT DESIGN (1.03)
     - RAISE the solidity x1.41 (stage L/g2 0.53 -> 0.75): R2 0.50 -> 0.70, S2 0.50 -> 0.70; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 18% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.95
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.97
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
```

Inputs: Max_eta_Compressor_design_rs_no_last_stator (RS)

Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_eta_Compressor_design_rs_...		
alpha1	30.3676	deg	.mat: Max_eta_Compressor_design_rs_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rs_...
A_vec	0.00283128 0.00262358 0.00243314 0....	m^2	.mat: Max_eta_Compressor_design_rs_...
beta1_dash	50.3693	deg	.mat: Max_eta_Compressor_design_rs_...
beta2_dash	18.077	deg	.mat: Max_eta_Compressor_design_rs_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rs_...
w_loss_rotor	0.0324827	-	.mat: Max_eta_Compressor_design_rs_...
deviation	12.2907	deg	.mat: Max_eta_Compressor_design_rs_...
incidence	-6.04865	deg	.mat: Max_eta_Compressor_design_rs_...
deflection	13.953	deg	.mat: Max_eta_Compressor_design_rs_...
w_loss_stator	0.0266822	-	.mat: Max_eta_Compressor_design_rs_...
U	274.775	m/s	.mat: Max_eta_Compressor_design_rs_...
sigma_rotor	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
sigma_stator	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rs_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

Diagnostics: Max_eta_Compressor_design_rs_no_last_stator (RS)

```
Rotor-Stator  |  PR_des = 1.4939, eta_des = 0.9312  |  correlations: polynomial, Howell factor 2.00

STABILITY (design speed): first limit = peak at m = 0.4809 kg/s (PR 1.554); stall margin SM = 8.2 %
   peak of the speed line (zero slope): m = 0.4809 kg/s | Howell stall onset: m = 0.4787 kg/s
   Koch stage limit: m = 0.5018 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5151 kg/s (Rotor 2 exit (relative)) -> choke margin 3.0 %, operating range (m_choke - m_lim)/m_choke = 6.6 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.377 -> surge if the installation has V_p/(A_c L_c) > 4.51 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 1 (omega/omega_des = 1.87), lowest de Haller Rotor 1 (0.79). Details: Tables > Blade
      rows.

KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: drawn on 9 of 15 speed lines (N/N_des 0.80 to 1.20).
   why: Stage 1 (R1+S1): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2 only): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): BEYOND THE LIMIT AT DESIGN (1.04)
     - RAISE the solidity x1.42 (stage L/g2 0.53 -> 0.76): R1 0.50 -> 0.71, S1 0.50 -> 0.71; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 18% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.95
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.97
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
  Stage 2 (R2 only): BEYOND THE LIMIT AT DESIGN (1.03)
     - RAISE the solidity x1.39 (stage L/g2 0.53 -> 0.74): R2 0.50 -> 0.70; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 17% at the same velocities (lower loading psi or degree of reaction, or a larger flow
      coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.94
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.96
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
```